

APSPDCL · COMMERCIAL ROOFTOP

400 KWP GRID-TIED SOLAR PLANT

Net investment ₹1.54 Cr after input tax credit, generating 6,13,200 units in year one against an energy charge of ₹7.65 per unit.

HT CATEGORY I / II AT 11 KV

30-YEAR ANALYSIS

PAYBACK

3.7 yrs

Post-tax, undiscounted

PROJECT IRR

27.5%

Unlevered, 30 years

COST OF SOLAR ENERGY

₹3.15

per kWh, against ₹7.65 from the grid

NET PRESENT VALUE

₹2.85 Cr

Discounted at 10%

PLANT AND GENERATION

Plant capacity	400 kWp
Year 1 generation	6,13,200 kWh
Specific yield	1,533 kWh / kWp / year
Supply category	HT Category I / II at 11 kV (HT supply at 11 kV)
Energy charge offset	₹7.65 / kWh
Units self-consumed	90% at the energy charge
Units exported	10% at ₹2.75 / kWh (APPC)
Year 1 gross saving	₹43.91 L

INVESTMENT AND TAX BASIS

Price per watt-peak	₹40.50 / Wp (incl. GST)
Project price incl. GST	₹1.62 Cr
GST at 5% — recovered as ITC	₹7.71 L
Net capitalised cost	₹1.54 Cr
Depreciation block	40% WDV, Appendix I
Tax regime	s.115BAA — 25.168%
Annual maintenance	₹600 / kWp / year, year 1 free
30-year depreciation shield	₹38.83 L

ACCELERATED DEPRECIATION — FIRST 12 YEARS

YEAR	OPENING WDV (₹)	DEPRECIATION AT 40% (₹)	CLOSING WDV (₹)	TAX SHIELD AT 25.168% (₹)
1	1,54,28,572	61,71,429	92,57,143	15,53,225
2	92,57,143	37,02,857	55,54,286	9,31,935
3	55,54,285	22,21,714	33,32,571	5,59,161
4	33,32,572	13,33,029	19,99,543	3,35,497
5	19,99,543	7,99,817	11,99,726	2,01,298
6	11,99,725	4,79,890	7,19,835	1,20,779
7	7,19,835	2,87,934	4,31,901	72,467
8	4,31,902	1,72,761	2,59,141	43,480
9	2,59,140	1,03,656	1,55,484	26,088
10	1,55,485	62,194	93,291	15,653
11	93,290	37,316	55,974	9,392
12	55,975	22,390	33,585	5,635
Tax shield, years 1-12				38,74,610

COMMERCIAL SOLAR — FINANCIAL ANALYSIS

30-year post-tax cash flow

400 kWp

PLANT CAPACITY

How this table works. Electricity savings reduce an operating expense, so they increase taxable profit and are taxed at 25.168%. Depreciation reduces the same taxable profit. Both are applied to one figure, so the tax column is a net inflow in the early years when depreciation exceeds savings — shown in green. Net cash flow is therefore gross saving, less maintenance, less that tax. The shaded row is the year the investment is recovered.

YEAR	GENERATION (KWH)	GROSS SAVING (₹)	AMC (₹)	DEPRECIATION (₹)	TAX (₹)	NET CASH FLOW (₹)	CUMULATIVE (₹)
1	6,13,200	43,90,512	—	61,71,429	(4,48,221)	48,38,733	48,38,733
2	6,10,134	45,43,302	2,40,000	37,02,857	1,51,120	41,52,182	89,90,915
3	6,07,083	47,01,409	2,40,000	22,21,714	5,63,686	38,97,722	1,28,88,637
4	6,04,048	48,65,018	2,40,000	13,33,029	8,28,528	37,96,490	1,66,85,127
5	6,01,028	50,34,320	2,40,000	7,99,817	10,05,337	37,88,984	2,04,74,111
6	5,98,023	52,09,515	2,40,000	4,79,890	11,29,949	38,39,566	2,43,13,677
7	5,95,032	53,90,806	2,40,000	2,87,934	12,23,888	39,26,918	2,82,40,595
8	5,92,057	55,78,406	2,40,000	1,72,761	13,00,090	40,38,316	3,22,78,911
9	5,89,097	57,72,534	2,40,000	1,03,656	13,66,340	41,66,194	3,64,45,105
10	5,86,151	59,73,419	2,40,000	62,194	14,27,334	43,06,085	4,07,51,190
11	5,83,221	61,81,294	2,40,000	37,316	14,85,913	44,55,381	4,52,06,571
12	5,80,305	63,96,403	2,40,000	22,390	15,43,808	46,12,594	4,98,19,165
13	5,77,403	66,18,997	2,40,000	13,434	16,02,085	47,76,912	5,45,96,077
14	5,74,516	68,49,338	2,40,000	8,060	16,61,410	49,47,929	5,95,44,006
15	5,71,644	70,87,695	2,40,000	4,836	17,22,211	51,25,485	6,46,69,491
16	5,68,785	73,34,347	2,40,000	2,902	17,84,775	53,09,572	6,99,79,063
17	5,65,941	75,89,583	2,40,000	1,741	18,49,305	55,00,278	7,54,79,341
18	5,63,112	78,53,700	2,40,000	1,045	19,15,953	56,97,747	8,11,77,088
19	5,60,296	81,27,009	2,40,000	627	19,84,845	59,02,164	8,70,79,252
20	5,57,495	84,09,829	2,40,000	376	20,56,088	61,13,741	9,31,92,993
21	5,54,707	87,02,491	2,40,000	226	21,29,783	63,32,708	9,95,25,701
22	5,51,934	90,05,337	2,40,000	135	22,06,026	65,59,311	10,60,85,012
23	5,49,174	93,18,723	2,40,000	81	22,84,913	67,93,811	11,28,78,823
24	5,46,428	96,43,015	2,40,000	49	23,66,538	70,36,476	11,99,15,299
25	5,43,696	99,78,592	2,40,000	29	24,51,001	72,87,590	12,72,02,889
26	5,40,977	1,03,25,847	2,40,000	18	25,38,401	75,47,445	13,47,50,334
27	5,38,273	1,06,85,186	2,40,000	11	26,28,842	78,16,344	14,25,66,678
28	5,35,581	1,10,57,031	2,40,000	6	27,22,429	80,94,602	15,06,61,280
29	5,32,903	1,14,41,815	2,40,000	4	28,19,272	83,82,543	15,90,43,823
30	5,30,239	1,18,39,990	2,40,000	2	29,19,485	86,80,505	16,77,24,328
Total	1,71,22,483	22,59,05,463	69,60,000	1,54,28,569	5,12,21,134	16,77,24,328	16,77,24,328

RETURN METRICS

Simple payback	3.7 yrs
Discounted payback	4.9 yrs
Project IRR, unlevered	27.5%
Net present value at 10%	₹2.85 Cr

OVER THE FULL TERM

Levelised cost of energy	₹3.15 / kWh
Cumulative return, 30 years	987%
Total net cash flow, after tax	₹16.77 Cr
30-year generation	1,71,22,483 kWh

COST OF A UNIT — SOLAR AGAINST THE GRID

Solar, levelised over 30 years		₹3.15
Grid energy charge today		₹7.65

Every unit the plant generates costs ₹3.15 across its life against ₹7.65 bought from APSPDCL today — and the grid figure reaches ₹23.86 by year 30 at 4% escalation, while the solar figure is fixed at the day of purchase. Over 30 years the plant generates 1,71,22,483 units, avoiding 14,040.4 tonnes of CO₂.

ASSUMPTIONS

- Tariff:** HT Category I / II at 11 kV at ₹7.65/kWh. Energy charge; ₹475/kVA/month demand charge excluded. APERC Retail Supply Tariff Order FY2025-26, continued for FY2026-27.
- Demand charges:** Excluded. Solar does not reduce contracted demand, so fixed and demand charges are unchanged by the plant.
- Tariff escalation:** 4% per year, applied to both the energy charge and the export rate.
- Generation:** 1,533 kWh per kWp in year one, degrading 0.5% per year.
- Unit split:** 90% self-consumed, 10% exported at ₹2.75/kWh. The plant is assumed to be sized to the site's daytime load.
- GST:** 5%, recovered in full as input tax credit. Figures are stated ex-GST. A buyer who cannot claim ITC should read the cost as ₹1.62 Cr.
- Depreciation:** 40% written-down value, Income Tax Act Appendix I. No additional depreciation under s.32(1)(ia), which s.115BAA forfeits.
- Tax:** 25.168% — s.115BAA at 22% plus 10% surcharge and 4% cess. Applied to savings and to depreciation alike.
- Loss set-off:** Early-year depreciation exceeds savings. The resulting loss is assumed to shelter the buyer's other business income in the same year.
- Commissioning:** Full first-year depreciation assumed, i.e. the plant is put to use on or before 30 September. Later commissioning halves the year-one charge, with the balance carried forward.
- Maintenance:** ₹600 per kWp per year from year 2, held flat. Year 1 is covered by the build.
- Inverter replacement:** Not modelled. A mid-life inverter replacement would reduce the figures shown.
- Financing:** None. All returns are unlevered — the return on the money invested, before any loan.
- Horizon:** 30 years, which runs beyond the module performance warranty.
- Discount rate:** 10% for net present value, discounted payback and levelised cost.
- Emissions:** Grid emission factor 0.82 kg CO₂ per kWh; tree equivalence at 45 trees per tonne is illustrative, not measured sequestration.

BASIS AND LIMITATIONS

This is an indicative financial analysis for a standard 400 kWp commercial rooftop plant. **It is not a quotation, an offer, or tax advice.** Pricing is subject to a site survey, roof condition, structure height, cable routing, evacuation distance and the APSPDCL sanction actually granted.

Tariffs, the depreciation block and the corporate tax rate are as we understand them at the date printed below and are subject to change by APERC and by the Finance Act. The tax treatment shown assumes a GST-registered company that has opted into s.115BAA and has sufficient other business income to absorb the early-year depreciation. **Have your chartered accountant confirm the tax position before you rely on it.**

Generation depends on site irradiance, shading, orientation, soiling and grid availability. Actual results will differ from this projection.